

```
In [50]: import numpy as np
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import MultinomialNB
from sklearn.svm import SVC
from sklearn import metrics
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.feature_extraction.text import CountVectorizer

from sklearn import tree
```

```
In [51]: data = pd.read_csv('emails.csv', header=None)
data=data.rename(columns = {0:'message',1:'label'})
data.head()
```

```
Out[51]:
```

	message	label
0	Subject: naturally irresistible your corporate...	1
1	Subject: the stock trading gunslinger fanny i...	1
2	Subject: unbelievable new homes made easy im ...	1
3	Subject: 4 color printing special request add...	1
4	Subject: do not have money , get software cds ...	1

```
In [52]: df = pd.read_csv('EnglishSpam.csv', sep="\t", header = None)
df=df.rename(columns = {0:'label',1:'message'})
print(df['label'].value_counts())
```

```
label
ham      4825
spam      747
Name: count, dtype: int64
```

```
In [53]: df = df[df['label'] == 'spam']
print(df['label'].value_counts())
```

```
label
spam      747
Name: count, dtype: int64
```

```
In [54]: df['label'] = df['label'].replace({
    'spam': 1,
    'ham': 0
})
```

```
In [55]: data['message'] = data['message'].str.replace(r'^Subject:\s*', '', regex=True)
data.head()
```

Out[55]:

	message	label
0	naturally irresistible your corporate identity...	1
1	the stock trading gunslinger fanny is merrill...	1
2	unbelievable new homes made easy im wanting t...	1
3	4 color printing special request additional i...	1
4	do not have money , get software cds from here...	1

In [56]: `data = pd.concat([data, df], ignore_index=True)`

In [57]: `print(data['label'].value_counts())`

```
label
0    4360
1     2115
Name: count, dtype: int64
```

In [58]: `data.to_csv("final_dataset.csv", index=False)`

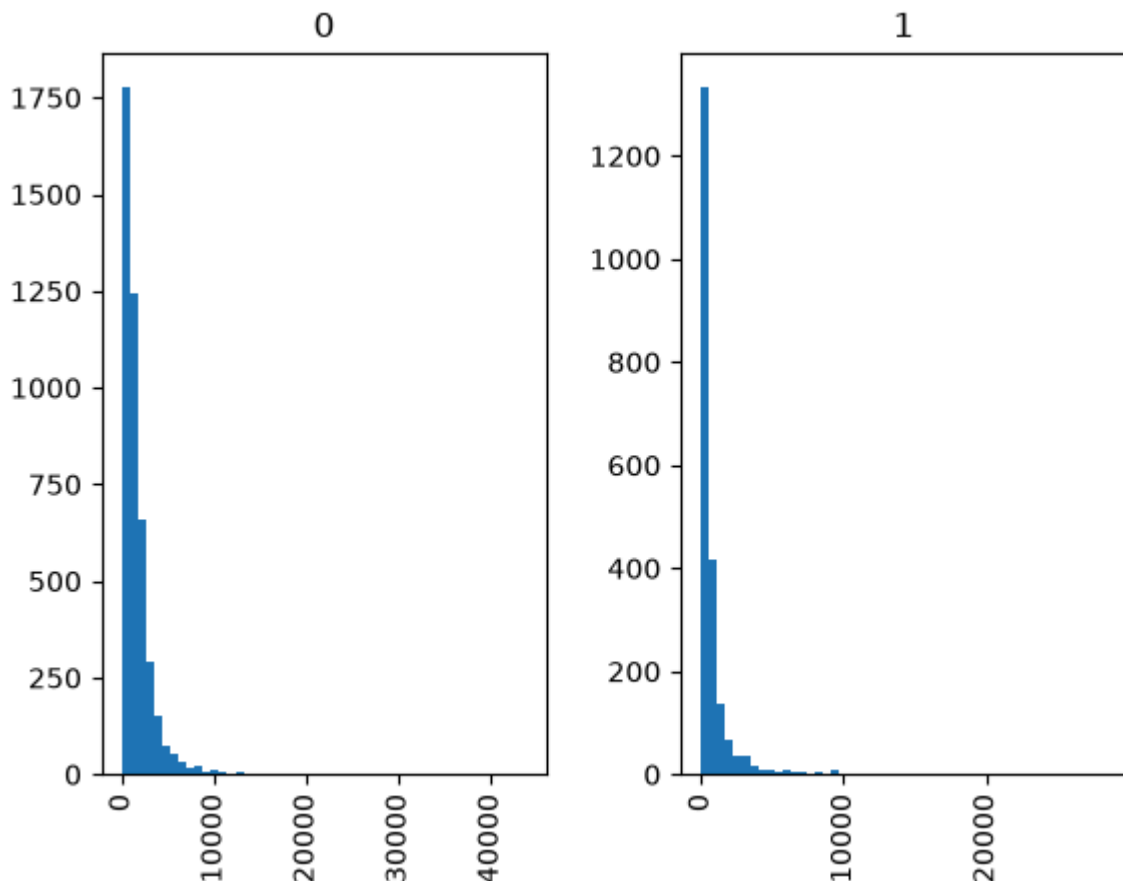
In [59]: `data=data.sample(frac=1)`
`data.head()`

Out[59]:

	message	label
5042	swaps monitor research . we have today publis...	0
2399	copier on 32 nd floor hello iain , well time...	0
3655	re : real options paul , krishna and i are t...	0
4348	re : message from bogdan thanks vince . we ' ...	0
4823	wallet size telephone cards for the research g...	0

In [60]: `data['length'] = data['message'].map(lambda text: len(text))`
`data.hist(column='length', by='label', bins=50)`

Out[60]: `array([<Axes: title={'center': '0'}>, <Axes: title={'center': '1'}>],
dtype=object)`



```
In [61]: # train test split
data['label'] = pd.to_numeric(data['label'])
x_train, x_test, y_train, y_test = train_test_split(data.message, data.label, te
```

```
In [62]: x_train
```

```
Out[62]: 3831    re : willow and pathstar evaluations  please r...
1099    big range of all types of downloadable softwar...
4479    re : i am zhendong  sure thing !  - - - - - or...
2758    re : trading algorithms  andy ,  sounds good ....
5378    re : thanksgiving staff meeting  hi clayton , ...

...

656     you launched a website but no one visits it ? ...
4771    re : weather and energy price data  mulong wan...
124     enhance your anatomy  the longz system , both ...
6141    FREE UNLIMITED HARDCORE PORN direct 2 your mob...
60      partnership  mr . edward moko  18 independence...
Name: message, Length: 4532, dtype: str
```

```
In [63]: data = pd.concat([data, df], ignore_index=True)
data['label'] = data['label'].astype(int)
```

```
In [64]: ## can also be used tfidf vectorizer instead of countvectorizer
cv = CountVectorizer()
x_train_count = cv.fit_transform(x_train.values)
```

```
In [65]: x_train_count
```

```
Out[65]: <Compressed Sparse Row sparse matrix of dtype 'int64'
         with 509671 stored elements and shape (4532, 32716)>
```

```
In [66]: x_train_count.toarray()
```

```
Out[66]: array([[0, 0, 0, ..., 0, 0, 0],
               [0, 0, 0, ..., 0, 0, 0],
               [0, 0, 0, ..., 0, 0, 0],
               ...,
               [0, 0, 0, ..., 0, 0, 0],
               [0, 0, 0, ..., 0, 0, 0],
               [4, 8, 0, ..., 0, 0, 0]], shape=(4532, 32716))
```

```
In [67]: # train model
model = MultinomialNB()
model.fit(x_train_count, y_train)
```

```
Out[67]: ▼ MultinomialNB ⓘ ?
          ► Parameters
          ► Fitted attributes
```

```
In [68]: #test model
x_test_count = cv.transform(x_test)
model.score(x_test_count, y_test)
```

```
Out[68]: 0.9927946474523932
```

```
In [69]: predicted=model.predict(x_test_count)
```

```
In [70]: predicted
```

```
Out[70]: array([0, 0, 0, ..., 0, 0, 0], shape=(1943,))
```

```
In [71]: print(metrics.classification_report(y_test,predicted))
```

	precision	recall	f1-score	support
0	0.99	1.00	0.99	1300
1	0.99	0.99	0.99	643
accuracy			0.99	1943
macro avg	0.99	0.99	0.99	1943
weighted avg	0.99	0.99	0.99	1943

```
In [72]: from sklearn.metrics import confusion_matrix

# Generate confusion matrix
cm = confusion_matrix(y_test, predicted, labels=[1, 0])

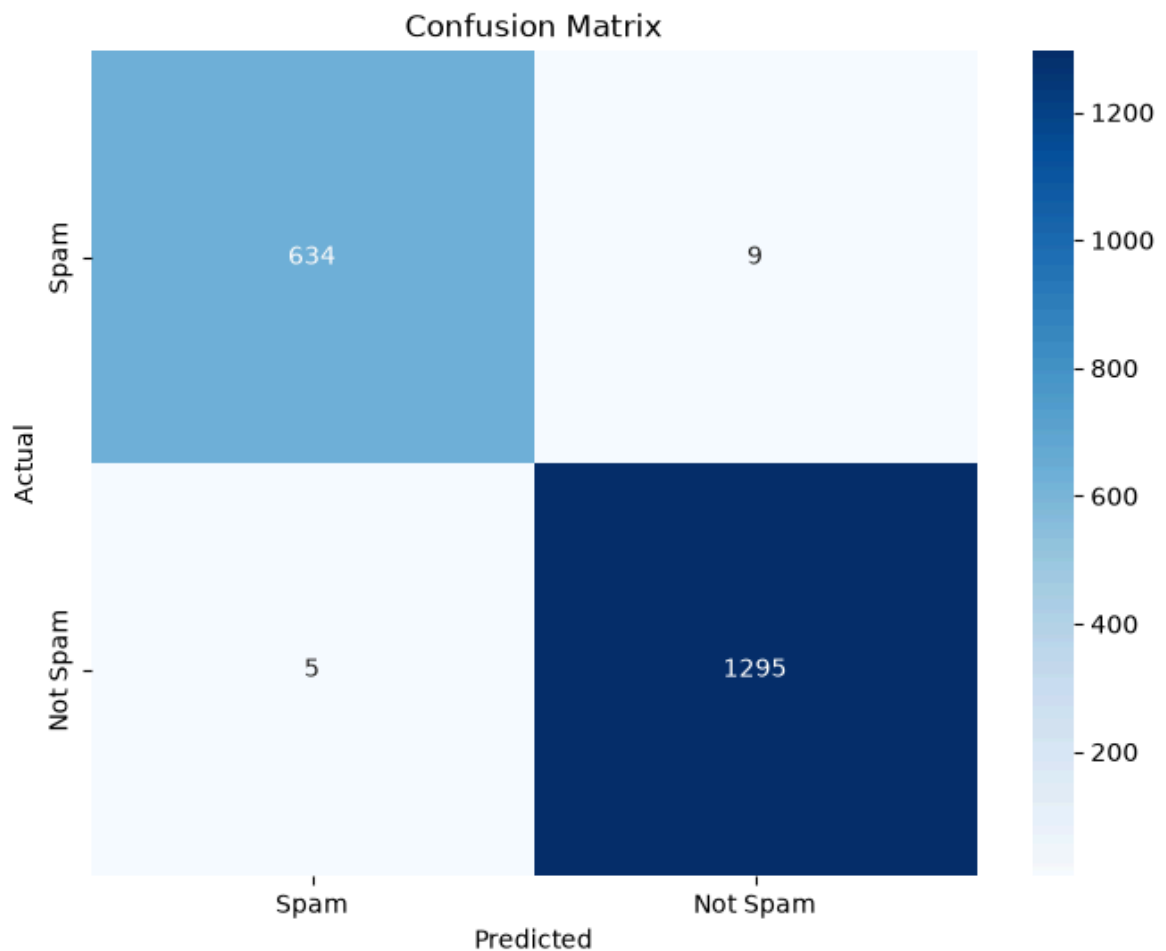
# Print confusion matrix
print("Confusion Matrix:")
print(cm)
```

```
import matplotlib.pyplot as plt
import seaborn as sns

plt.figure(figsize=(8, 6))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues', xticklabels=['Spam', 'Not Spa
plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.title('Confusion Matrix')
plt.show()
```

Confusion Matrix:

```
[[ 634   9]
 [   5 1295]]
```



Test the Result with different email.

```
In [73]: sample_emails = [
    "Hi, are we still meeting for lunch at 1 pm today?",
    "Your electricity bill is due tomorrow. Please use the official payment port",
    "Congratulations! You have won $1,000,000. Call now to claim your prize.",
    "URGENT: You have been selected for a free vacation. Text WIN to 80000 now!",
    "Can you send me the notes from today's class?",
    "Free entry in a weekly competition to win a cash prize. Reply WIN now!"
]

sample_email_count = cv.transform(sample_emails)
sample_predictions = model.predict(sample_email_count)

pd.DataFrame({
    "email": sample_emails,
```

```
"prediction": ["spam" if prediction == 1 else "ham" for prediction in sample  
})
```

Out[73]:

	email	prediction
0	Hi, are we still meeting for lunch at 1 pm today?	ham
1	Your electricity bill is due tomorrow. Please ...	ham
2	Congratulations! You have won \$1,000,000. Call...	spam
3	URGENT: You have been selected for a free vaca...	spam
4	Can you send me the notes from today's class?	ham
5	Free entry in a weekly competition to win a ca...	spam